Multilevel Fair Allocation: on fair allocation in hierarchical structures

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Fair Allocation : motivations

Fair allocation deals with the design of algorithms to allocate fairly some resources.

Traditionally, it focused on fairness between individuals, e.g. divorce settlements or inheritance.

More recently, fair allocation between groups of individuals drew interest, e.g. allocating office spaces to teams of a laboratory.

Classical Fair Allocation

Fair Allocation : model

A simple model

- a set of agents $\mathcal{N} = \{1, \ldots, n\}$
- a set of indivisible resources $\mathcal{G} = \{g_1, \ldots, g_m\}$
- an agent $i \in \mathcal{N}$ has an additive utility function over $\mathcal{G}$, s.t. $v_i(S) = \sum_{g \in S} v_i(\{g\})$ for $S \subseteq \mathcal{G}$

Allocation

$A : \mathcal{N} \to 2^{\mathcal{G}}$ is an allocation if

- $\cup_{i \in \mathcal{N}} A(i) \subseteq \mathcal{G}$
- $A(i) \cap A(j) = \emptyset$ for any $i, j \in \mathcal{N}$

Fair Allocation : fairness

Researchers have proposed different approaches to fairness :

- Optimization-based fairness
  - maxmin : maximizing the lowest utility
  - leximin : maximizing the lowest utility, subject to that, maximizing the second lowest utility, and so on.
- Axiomatic fairness

A popular notion of fairness is envy-freeness.

Envy-freeness (EF)

Allocation A is envy-free if for any pair of agents $i, j \in \mathcal{N}$, we have

$$v_i(A(i)) \geq v_i(A(j))$$

$\Rightarrow$ an EF allocation is not guaranteed to exist : suppose 2 agents, and 1 item.

Envy-freeness up-to-one good (EF1)

Allocation A is envy-free up-to-one good if for any pair of agents $i, j \in \mathcal{N}$, we have

$$v_i(A(i)) \geq v_i(A(j) \setminus \{g\})$$

for some $g \in A(j)$.

Fair Allocation : round robin algorithm

The simplest algorithm to compute an EF1 allocation is the Round Robin picking sequence.

Round Robin algorithm

- Fix a sequence $1, 2, \ldots, n, 1, 2 \ldots$ of length m.
- Iterate over this sequence, letting each agent pick her favorite item among the remaining ones.

Multilevel Fair Allocation

Multilevel Fair Allocation : motivations

Classical fair allocation is great, but in real-life, agents may belong to groups, which themselves belong to higher-level groups, ultimately forming a hierarchical structure.

- Allocation of resources across a territory
- Allocation of materials in a hierarchical organization, e.g. a university
- Allocation of tasks to robots with organizational roles [Guillet et al., 2026]

Multilevel Fair Allocation under Additive Preferences

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9th *International Conference on Algorithmic Decision Theory (ADT 2026)*

Paper : [Lucet et al., 2026]

Multilevel Fair Allocation : motivations

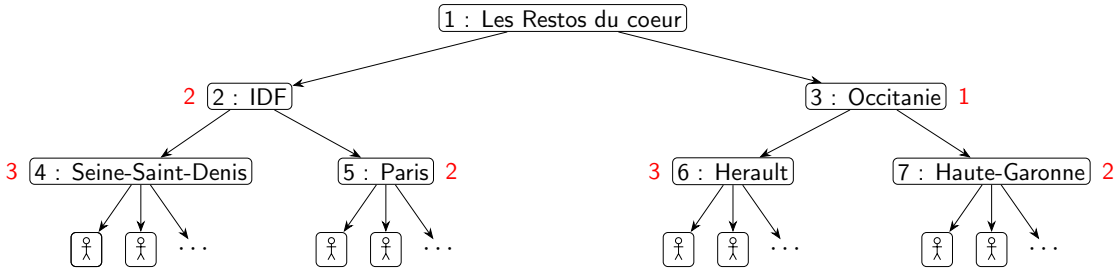


Figure – Simplified structure of Les Restos du coeur.

Multilevel Fair Allocation

We introduce a fair division model within a hierarchical structure, called **multilevel fair allocation**.

Multilevel Fair Allocation : Model

Notations

- a set of entities $\mathcal{N} = \{1, \ldots, n\}$, structured as a tree;
- each node $i \in \mathcal{N}$ has a weight w_i;
- a set of items $\mathcal{G} = \{g_1, \ldots, g_m\}$ to allocate.

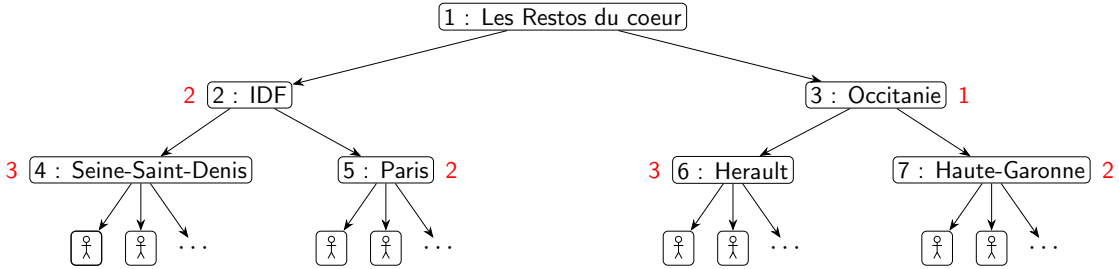


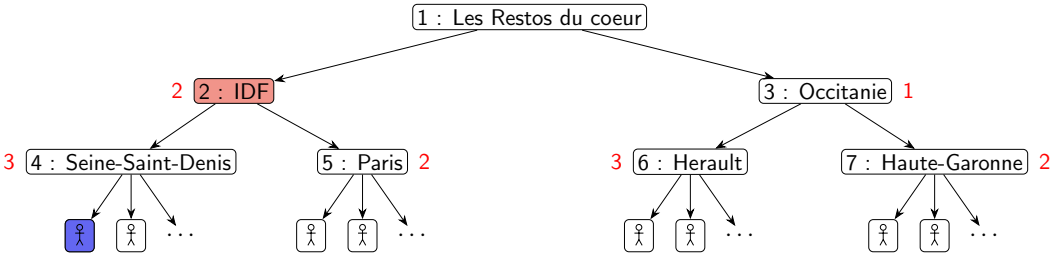
Figure – Simplified structure of Les Restos du coeur.

Definition

A **multilevel allocation** $\pi : \mathcal{N} \to 2^{\mathcal{G}}$ maps any node to a bundle of items such that :

- an internal node receives items only to allocate it to its children.
- given two siblings i, j, we have $\pi(i) \cap \pi(j) = \emptyset$.

We denote $\pi|_{\text{children}(i)}$ the local allocation of i's bundle to its children.



Internal node's utility

The utility v_i of an internal node i is the sum of the utility of its children, i.e.

$$v_i(\pi) = \sum_{j \in \text{children}(i)} v_j(\pi)$$

Leaf's utility

The utility u_x of a leaf x is additive "over the items", i.e.

$$u_x(S) = \sum_{g \in S} u_x(g)$$

Multilevel Fairness

A multilevel allocation π is multilevel fair if, for any internal node i, the allocation $\pi|_{\text{children}(i)}$ is fair.

WEF : weighted envy-freeness

For any pair of agents i, j,

$$\frac{i\text{'s utility}}{i\text{'s weight}} \geq \frac{i\text{'s utility for } j\text{'s bundle}}{j\text{'s weight}}$$

WEF : weighted envy-freeness

For any pair of agents i, j,

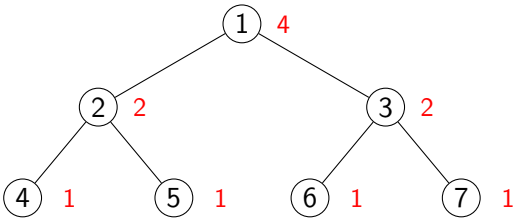
$$\frac{i\text{'s utility}}{i\text{'s weight}} \geq \frac{i\text{'s utility for } j\text{'s bundle}}{j\text{'s weight}}$$

$\Rightarrow$ **Problem** : how does an internal node evaluate the bundle of one of its siblings' bundle?

We assume internal nodes will estimate the value it could derive from a sibling's bundle :

- Optimistic utility $\hat{v}$: computes a *utilitarian-optimal* allocation to its leaves, and measures the utilitarian welfare.
- Pessimistic utility $\overset{\vee}{v}$: computes a *utilitarian-pessimal* allocation to its leaves, and measures the utilitarian welfare.
- Agnostic utility $\overline{v}$: computes the weighted average utility of the bundle to its leaves.

Example



Objets	g_1	g_2	g_3	g_4	g_5
4	2	2	2	2	2
5	1	1	1	1	1
6	2	2	2	2	2
7	1	1	1	1	1

Table – Leaves' preferences

Example of allocation :

$$\left.\begin{array}{c}\pi(4)=\{g_1,g_5\}\\ \pi(5)=\{g_3\}\end{array}\right\}\pi(2)=\{g_1,g_3,g_5\}$$

$$\left.\begin{array}{c}\pi(6)=\{g_2\}\\ \pi(7)=\{g_4\}\end{array}\right\}\pi(3)=\{g_2,g_4\}$$

Estimated utilities

Pessimistic : $\overset{\vee}{v}_3(\{g_1,g_3,g_5\})=3$

Agnostic : $\overline{v}_3(\{g_1,g_3,g_5\})=4.5$

Optimistic : $\hat{v}_3(\{g_1,g_3,g_5\})=6$

Definition

Multilevel allocation π is *estimated multilevel weighted envy-free up-to-one good* for $\mathcal{E} \in \{\text{pess}, \text{agno}, \text{opt}\}$ (denoted by M[$\mathcal{E}$]-WEF1), if for any internal node i, and any pair of its children j, k, we have

$$\frac{v_j(\pi)}{w_j} \geq \frac{\tilde{v}_j(\pi(k) \setminus \{g\})}{w_k}$$

for some $g \in \pi(k)$, where $\tilde{v}$ denotes $\overset{\vee}{v}$, $\overline{v}$ or $\hat{v}$ for $\mathcal{E} = \text{pess}$, agno, or opt, respectively.

Multilevel Fair Allocation : Theoretical and experimental results

Multilevel Fair Allocation : Multilevel Weighted Round Robin

Multilevel extension of the Weighted Round Robin [Chakraborty et al., 2021].

Algorithm

At every iteration :

- starting at the root, pick recursively a node that is the least picked child relative to its weight, until a leaf was selected.
- Let this leaf pick her favorite remaining item.
- Repeat until all items are allocated.

Multilevel Fair Allocation : MWRR guarantees M[pess]-WEF1

Theorem

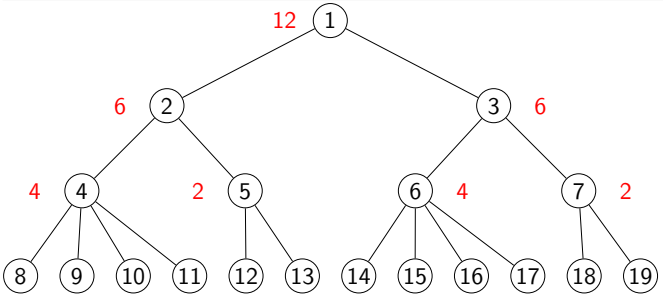
MWRR always returns an M[pess]-WEF1 allocation.

Remark : we show that under identical valuations all three notions coincide, and hence MWRR guarantees our three notions.

Multilevel Fair Allocation : M[agno]-WEF1 is not guaranteed to exist

Theorem

Under general additive valuations, an M[agno]-WEF1 allocation needs not exist.



Item	g_1	g_2	g_3	g_4	g_5
C(4)	2	2	2	2	2
C(5)	1	1	1	1	1
C(6)	2	2	2	2	2
C(7)	1	1	1	1	1

Table – Leaves' preferences over items

Multilevel Fair Allocation : MWRR does not guarantee M[agno]-WEF1

Theorem

Under general additive valuations, MWRR does not guarantee an M[agno]-WEF1 allocation, even when it exists.

$\Rightarrow$ It cannot even be approximated!

Corollary

Under general additive valuations, MWRR does not guarantee an M[opt]-WEF1 allocation, even when it exists.

Multilevel Fair Allocation : Beyond worst-case with experimental tests

Experimental protocol :

- three types of trees : balanced, comb, partially unbalanced.
- either random weights, or dependent on the number of leaves.
- preferences generated with different techniques : mallows, resampling-dirichlet, cost utilities, correlated preferences.

Results

- Out of 200,000 generated instances, only 35 allocations returned by MWRR were not M[agno]-WEF1.
- M[opt]-WEF1 is much more challenging : in some settings, close to 100% allocations returned were not fair.

Perspectives

Perspectives

In this paper, we studied a fair allocation problem within a hierarchical structure. We showed that :

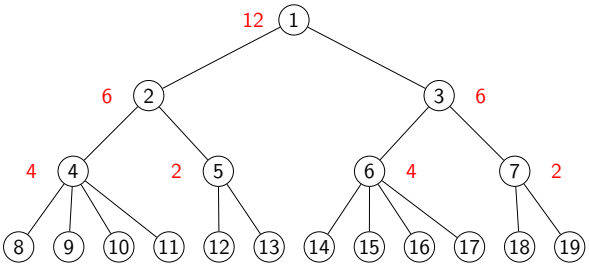
- Adapting fairness notions to the multilevel setting is **not neutral**.
- A multilevel extension of the WRR could guarantee the weakest fairness notion, but not the stronger ones – but behaved very well experimentally.

Open questions and future directions

- Is M[agno]-WEF1 guaranteed to exist under normalized utilities?
- What if the hierarchical organization is a DAG instead of a tree?

Thanks for your attention

MWRR does not guarantee M[agno]-WEF1



Objets	g_1	g_2	g_3	g_4	g_5
C(4)	5/21	5/21	1/21	5/21	5/21
C(5)	1/5	1/5	1/5	1/5	1/5
C(6)	5/21	5/21	5/21	1/21	5/21
C(7)	1/5	1/5	1/5	1/5	1/5

Table – Leaves' preferences

Returned allocation :

- $\pi(8) = \{g_1\}$
- $\pi(9) = \{g_5\}$
- $\pi(12) = \{g_3\}$
- $\pi(14) = \{g_2\}$
- $\pi(18) = \{g_4\}$

Non-M[agno]-WEF1 allocation

$$v_3(\pi) = 0.438$$

$$\bar{v}_3(\pi(2) \setminus \{g_1\}) = 0.451$$

Bibliography


Chakraborty, M., Igarashi, A., Suksompong, W., and Zick, Y. (2021).
Weighted envy-freeness in indivisible item allocation.
ACM Trans. Econ. Comput., 9(3).


Guillet, V., Grand, C., Lesire, C., and Picard, G. (2026).
Solving organizational multi-robot task allocation problems with consensus-based auctions.
In *2nd International Workshop on Autonomous Agents and Multi-Agent Systems for Space Applications (MASSpace-26)*, Paphos, Cyprus.


Lucet, M., Benabbou, N., Beynier, A., and Maudet, N. (2026).
Multilevel fair allocation under additive preferences.